

SCORE: TOKENIZATION-AWARE TRANSFORMER FOR SYMBOLIC MUSIC GENERATION WITH EXPRESSIONS

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ABSTRACT

Ever since the electrification of guitar, there has been an explosion in rhythmic complexity, tone experimentation and playing techniques. In this paper, a novel tokenisation method is proposed, which encapsulates the various guitar techniques and a sequence level attention component for the transformer architecture, which captures segmentation between musical phrases. The given approach enables efficient batching of variable length musical phrases while preserving their contextual coherence.

1 INTRODUCTION

A *riff* is a short, repeated motif or figure in the melody or accompaniment of a musical composition (Randel, 1986). This type of composition is particularly prevalent in rock, metal, blues, funk and jazz. It is often the main feature that gives character and structure to a song. Developments in Transformers architecture have inspired many to generate sequences by treating musical notes as symbolic language (Huang et al., 2018) (Agostinelli et al., 2023) (Copet et al., 2024) (Cui et al., 2025) (Muhammed et al., 2021) (Ferreira et al., 2023) (Huang & Yang, 2020a) (Wu & Yang, 2020) (Ens & Pasquier, 2020) (Dong et al., 2023) (Yu et al., 2022). Generating modern music demands capturing not only note patterns but also the associated expressions.

A widely adopted method for representing musical notes digitally is the MIDI format (Craig, 2025), which encodes pitch, duration, and dynamics. Using such a format for musical expression has limitations like resolution and bandwidth constraints (Shamoon, 2024) when compared to richer representations like audio recordings and detailed scores. Modern guitar riffs frequently employ intense and abrupt expressive techniques, which contribute significantly to its overall identity (Vallejo, 2021). To accurately represent the nuanced expressive elements of modern music, a score based format is preferable over simpler encodings like MIDI. The proposed method requires a tokenisation scheme that represents notes, beats, beat types and their associated accents, necessitating a dataset that includes expressive performance details.

In the context of symbolic music, notes are represented as sequences of discrete tokens. This requires a tokenisation scheme which directly affects how effectively a model can learn and generate coherent musical structures. Most existing approaches rely on methods similar to text processing such as BPE (Byte Pair Encoding) (Sennrich et al., 2016), WordPiece (Schuster & Nakajima, 2012) and SentencePiece (Kudo & Richardson, 2018). Although subword tokenization improves the structure (Kumar & Sarmiento, 2023) (Fradet et al., 2023), it becomes limited when dealing with the expressive element combinations of modern music. Recent studies (Gardín, 2024) have explored modeling chords based on scale degree, tonality and octave, and notes encoded relative to the current chord. This approach, however, is unable to produce every possible chord, particularly when dissonance is involved, which is becoming increasingly popular in modern metal. Most prior studies use absolute pitch values (Suresh et al., 2020), some (Kermarec et al., 2022) investigate the use of pitch intervals. This introduces an additional processing step, increasing inference time. Another common strategy treats each unique chord as a separate token, resulting in a large vocabulary that may include thousands of tokens representing all possible chords in the dataset (Dalmazzo et al., 2024b) (Dalmazzo

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Table 1: Beat type. $d = 7$ for dotted note, 0 otherwise.

b_t	Value
Triplet(3)	$1 + d$
Quintuplet(5)	$2 + d$
Sextuplet(6)	$3 + d$
Septuplet(7)	$4 + d$
Nonuplet(9)	$5 + d$
Undecuplet(11)	$6 + d$

Table 2: Beats.

b_b	Value
1 / 64	0
1 / 32	1
1 / 16	2
1 / 8	3
1 / 4	4
1 / 2	5
1	6

et al., 2024a). (Chen et al., 2020) proposes a pre-defined set of rhythmic groove patterns. This limits the model’s ability to generate novel rhythms.

Despite the growing body of work on symbolic music generation, there is little to no prior work specifically addressing expressive guitar music that captures techniques such as bends, slides, and chords in a tokenized representation. To address these limitations, a tokenisation scheme is proposed, that captures single and multiple musical notes efficiently and their associated accentuated elements. Furthermore, a sequence level attention mechanism is introduced to capture the segmentation between different musical phrases, facilitating batching of variable length phrases while maintaining musical context. The modified architecture is trained and evaluated on a dataset (Begari, 2025a) that includes guitar specific techniques.

While direct comparison with existing methods is not feasible due to differences in model compatibility and expressive feature representation, the proposed method is evaluated using quantitative metrics. The model is trained to minimize KL divergence across pitch, beat, beat type, and expression similarity. The source code of this work is available at <https://github.com/DjgentleViBe/SCORE>.

2 METHODS

2.1 TOKENISATION

2.1.1 NOTE REPRESENTATION

In place of absolute pitch indices, both fret and string numbers are employed to represent notes. This approach allows for a more compact representation of notes, as the same pitch can be played on different strings and frets. This is particularly relevant to guitar music, where notes can be produced at multiple fretboard positions, often resembling chord shapes. The model focuses on learning the relative motion of notes across frets, emphasizing positional changes rather than absolute pitch representation. This provides flexibility to accommodate alternative tunings (Weissman, 2006). The formula for note encodings for fretted string instrument is given by:

$$n_e = n_i + 2b_t \cdot n_s \cdot n_{ps} + 2b_b \cdot n_s \cdot n_{ps} \quad (1)$$

where, n_s = number of strings, n_{ps} = number of pitch steps per string.

$$n_i = f_n + s_n \cdot n_{ps} + n_s \cdot n_{ps} \cdot n_{val} \quad (2)$$

where, f_n = fret number, s_n = string number and $n_{val} = 1$ for palm mute, 0 otherwise; b_t and b_b are defined in 1 and 2 respectively.

2.1.2 ACCENT REPRESENTATION

A description of expressive elements is provided in (Begari, 2025b). An accent token is append to the note token from Section 2.1.1 to indicate the presence of an expressive element. For chords, the accent token is added before the next consecutive note token. A C major chord as shown in Figure 1 comprises of 3 notes - root note (C) played on the fret 2, 5th string, major third (G) on fret 1, 4th string and perfect fifth (E) on fret 2, 1st string (Alexander, 2019). The tokenization for the chord is

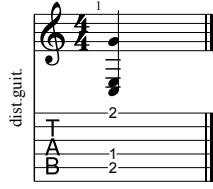


Figure 1: C major chord shape on a 6 string. The diagram illustrates musical notation along with guitar tablature.

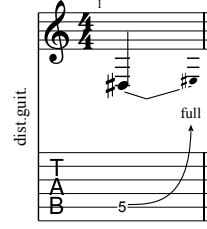


Figure 2: Musical notation and guitar tablature showing full note bend on fret 5, string 5.

as follows: 15413, 27051, 15389, 27051, 15321, 27051. Here, 27051 is the chord accent token, `BARRE_NOTEA`. Similarly, for a bend note depicted in Figure 2, the tokenization is: 15461, 27052. Here, 27052 is the bend accent token, `BEND_NOTE_1A`.

2.1.3 PHRASES

A music bar (also called a measure) is a fundamental unit in musical notation that divides music into segments of equal time, defined by the time signature (Miller, 2016). Each measure is separated by a `BARA`.

2.2 EMBEDDING

A bar consists of a fixed number of beats, composed of musical notes that were tokenised as described in Section 2.1. Each bar is treated like a sentence. It has been demonstrated that the transformer equipped with relative attention is very well-suited for generative modelling of symbolic music (Huang et al., 2018) (Zhou et al., 2023) (Li et al., 2024). The chunking strategy used to carry over context between sequences during training follows the approach introduced in (Yang et al., 2019), which builds upon Transformer-XL’s (Dai et al., 2019b) segment-level recurrence mechanism. This method enables the model to capture longer-range dependencies by reusing hidden states from previous token chunks across segments. An example of a riff is shown in is given in Figure 3.

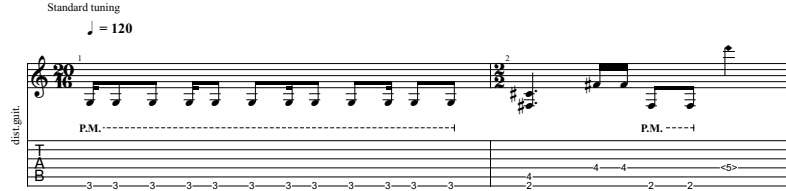


Figure 3: Example guitar riff showing a sequence of notes in combination with accents like pinch harmonic, barre chord and palm mute.

An example 16 tokens per sequence embedding is as follows: [27049, 7985, 11849, 11849, 7985, 11849, 11849, 7985, 11849, 11849, 7985, 11849, 11849, 27049, 17485, 17506]

[11849, 11849, 7985, 11849, 11849, 27049, 17485, 17506, 27051, 7802, 7802, 11848, 11848, 15530, 27074, 27049]

[27051, 7802, 7802, 11848, 11848, 15530, 27074, 27049, 27075, 0, 0, 0, 0, 0, 0]

The first two sequences contain musical notes, while the third sequence is padded with zeros to maintain a consistent length of 16 tokens. This padding ensures that all input sequences are of uniform length, while still allowing the flexibility to handle variable-length musical phrases.

2.3 TRAINING

2.3.1 DATASET

The guitar tracks were obtained from (Begari, 2025a). The dataset consists of files in .gp5 (Abakumov, 2025) format in standard tuning. From the full set, 6208 sequences were randomly sampled to enable efficient training on standard desktop hardware. The common practice of 80/20 split was used to create the training and validation sets (Goodfellow et al., 2016). For testing, top 10 first notes of each measure in the training set were selected as prompts for sequence generation and evaluation. Relevant statistics about the dataset used for training and evaluation are provided in (Begari, 2025b).

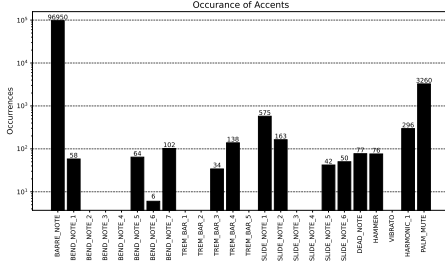


Figure 4: Distribution of accent tokens in the training data after random sampling, highlighting common expressive elements that the model learns to reproduce.

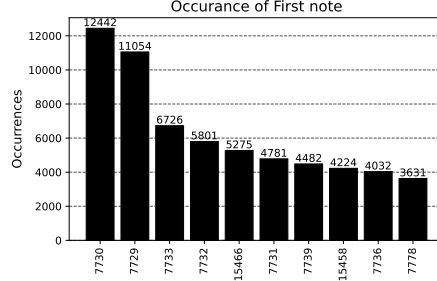


Figure 5: Distribution of starting notes for musical measures, used to select initial tokens for more consistent generation.

2.3.2 MODEL

The Transformer is built from a decoder-only stack with masked self attention as described in (Huang et al., 2018). Although, segment-level recurrence mechanism is captured in Transformer-XL (Dai et al., 2019b), its memory is limited to one previous segment at a time. To model dependencies across a whole batch, the gaussian distributions of the sequences are learned through a Multi layer Perceptron layer. If the input is a batch of sequences defined by:

$$X \in \mathbb{R}^{B \times T \times d} \quad (3)$$

where:

- B : batch size (number of sequences)
- T : number of tokens per sequence
- d : embedding dimension
- $X_i \in \mathbb{R}^{T \times d}$: the i -th sequence in the batch

Sequence level mean pooling is performed by:

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^T X_{i,t} \in \mathbb{R}^d \quad (4)$$

Gaussian parameters are then computed with a Multi Layer Perceptron. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^{2d}$ be a learned MLP. Then:

$$[\mu_i, \log \sigma_i] = f(\bar{x}_i) \quad \text{with} \quad \mu_i, \log \sigma_i \in \mathbb{R}^d \quad (5)$$

Ensure positivity of standard deviation:

$$\sigma_i = \exp(\log \sigma_i) + \varepsilon \quad \text{where} \quad \varepsilon > 0 \quad (6)$$

Each sequence is then represented as a diagonal multivariate Gaussian:

$$\mathcal{N}_i = \mathcal{N}(\mu_i, \text{diag}(\sigma_i^2)) \quad (7)$$

The pairwise KL divergence between each pair of Gaussians (Zhang et al., 2023):

$$\text{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) = \frac{1}{2} \sum_{k=1}^d \left[\log \left(\frac{\sigma_{j,k}^2}{\sigma_{i,k}^2} \right) + \frac{\sigma_{i,k}^2}{\sigma_{j,k}^2} + \frac{(\mu_{i,k} - \mu_{j,k})^2}{\sigma_{j,k}^2} - 1 \right] \quad (8)$$

This produces a KL matrix:

$$\text{KL_matrix} \in \mathbb{R}^{B \times B}, \quad [\text{KL_matrix}]_{i,j} = \text{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) \quad (9)$$

A simpler version of the attention matrix from (Dai et al., 2019a) is used from which the KL divergence-based sequence bias is subtracted:

$$\text{RA}_{\text{KL}} = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + QR^\top - \alpha \cdot \text{KL}_{\text{bias}} \right) \quad (10)$$

A weighted sum over relative positional value contributions V_{rel} is then added:

$$\text{Output} = \text{RA}_{\text{KL}} \times V + \text{RA}_{\text{KL}} \times V_{\text{rel}} \quad (11)$$

2.3.3 LOSS

The model’s loss is composed of three components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{rep}} + \mathcal{L}_{\text{seq}} \quad (12)$$

Cross Entropy loss ($\mathcal{L}_{\text{CE}}$) Cross-entropy loss is the standard choice for training deep neural networks and logistic regression models for both binary and multi-class classification tasks. (ml glossary, 2017)

Repetition loss ($\mathcal{L}_{\text{rep}}$) Repetition helps organise musical ideas, making pieces more coherent and memorable. It gives listeners reference points and expectations, aiding in the recognition and recall of themes or motifs. (Hu et al., 2022) Composers have long used stochastic (probability-based) methods to inject randomness into repeated motifs, sometimes even employing quantum randomness for truly unpredictable results. (Yamada et al., 2021) This demands careful consideration, as it relies on the listener’s inherent ability to recognise patterns and their natural inclination toward novelty. Since it is already known that accent tokens^A cannot repeat, it is penalised through repetition loss, making them less likely to be selected again in subsequent steps. A variant of the repetition penalty introduced in (Keskar et al., 2019) is employed; however, instead of greedy decoding, multinomial sampling is used to generate tokens to get semantically different output sequences (Song et al., 2024). Including this loss during training removes the need for explicit constraints during inference for generating valid sequences.

$$\text{Let } p_{b,t,v} = \text{softmax} \left(\frac{\text{logits}_{b,t,v}}{T} \right) \quad (13)$$

for $b = 1, \dots, B$; $t = 0, \dots, T - 1$; $v = 1, \dots, V$

Compute average past probabilities:

$$\bar{p}_{b,t,v} = \frac{1}{t} \sum_{k=0}^{t-1} p_{b,k,v}, \quad \text{for } t \geq 1 \quad (14)$$

Repetition score (dot product of current and average previous probabilities)

$$R_{b,t} = \sum_{v=1}^V p_{b,t,v} \cdot \bar{p}_{b,t,v}, \quad \text{for } t \geq 1 \quad (15)$$

Final penalty (mean repetition score across batch and sequence, excluding)

$$\mathcal{L}_{\text{rep}} = \frac{\lambda}{B(T-1)} \sum_{b=1}^B \sum_{t=1}^{T-1} R_{b,t} \quad (16)$$

where λ is a repetition penalty weight hyper parameter.

Sequence loss ($\mathcal{L}_{\text{seq}}$) Let $\mathbf{H} \in \mathbb{R}^{B \times H \times T \times d}$ denote the hidden-state tensor produced by the model for a batch of B sequences, where H is the number of attention heads, T is the sequence length, and d is the per-head hidden dimensionality. The heads are first merged into a single model embedding of dimension $D = H \cdot d$, then mean-pool is taken over the temporal dimension. A per-sequence memory vector $\mathbf{m} \in \mathbb{R}^{B \times D}$ is added to obtain the final sequence embedding.

$$\tilde{\mathbf{H}}_b = \frac{1}{T} \sum_{t=1}^T \text{reshape}(\mathbf{H}_{b,:,t,:}) \in \mathbb{R}^D, \quad b = 1, \dots, B \quad (17)$$

$$\mathbf{E}_b = \tilde{\mathbf{H}}_b + \mathbf{m}_b \in \mathbb{R}^D, \quad b = 1, \dots, B \quad (18)$$

where $\mathbf{E} \in \mathbb{R}^{B \times D}$ is the batch of final sequence embeddings. To encourage structural consistency of the embedding space across successive training steps, a relative distance preservation loss is introduced. Let $\mathbf{E}^{(t)}$ and $\mathbf{E}^{(t-1)}$ denote the sequence embeddings at the current and previous steps, respectively. For each batch, the pairwise squared Euclidean distance matrices Dokmanic et al. (2015) is computed:

$$\mathbf{D}_{ij}^{(t)} = \left\| \mathbf{E}_i^{(t)} - \mathbf{E}_j^{(t)} \right\|_2^2, \quad \mathbf{D}_{ij}^{(t-1)} = \left\| \mathbf{E}_i^{(t-1)} - \mathbf{E}_j^{(t-1)} \right\|_2^2, \quad (19)$$

for all $i, j \in \{1, \dots, B\}$.

The regularization term is an implementation of relational distance preservation objectives used in similarity preserving knowledge distillation Tung & Mori (2019). Unlike prior teacher-student formulations, the model’s previous embedding state is used as a fixed reference, enforcing temporal consistency via a detached pairwise distance target. The relative distance regularization term is defined as:

$$\mathcal{L}_{\text{rel}} = \delta \cdot \frac{1}{B^2} \sum_{i=1}^B \sum_{j=1}^B \left(\mathbf{D}_{ij}^{(t)} - \text{stopgrad}(\mathbf{D}_{ij}^{(t-1)}) \right)^2, \quad (20)$$

where δ is a scalar weighting coefficient and $\text{stopgrad}(\cdot)$ denotes gradient detachment, making it a fixed reference geometry.

This loss constrains the evolution of the embedding space by preserving pairwise relationships between samples, rather than enforcing absolute alignment of embeddings. As a result, it promotes temporal stability and mitigates representational drift while allowing global transformations of the embedding space.

3 RESULTS

The idea of context carryover has been well established in (Dauphin et al., 2016) (Wei et al., 2021) (Chen et al., 2025) (Rae et al., 2019) (Beltagy et al., 2020) (Zaheer et al., 2021). During inference, the model carries over context from previous sequences by using the last token of the preceding measure as the first token for the subsequent measure, following the approach described in (Huang & Yang, 2020b). To compare the quality of predicted tokens (P) from true distributions (Q), relative entropy is calculated using Kullback-Leibler divergence, following the methodology in (Seo et al., 2016) (Zhang et al., 2025) (Spiliopoulou & Storkey, 2011):

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (21)$$

3.1 GRID SEARCH

To determine the optimal configuration, a grid search was performed over a predefined set of hyperparameter values. The search included combinations of key parameters such as learning rate, sequence length, model dimensionality, Each configuration was trained for 10 epochs, and performance was evaluated on a validation set using loss. The best-performing set of hyperparameters was selected based on the lowest validation score.

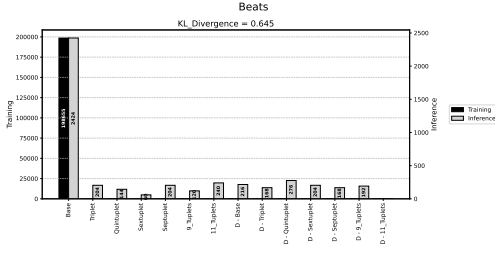


Figure 8: KL divergence for beat tokens with start token 7729, showing the model preserves rhythmic structure.

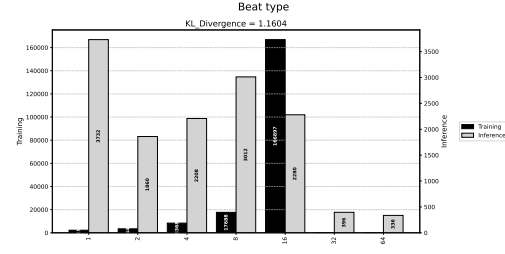


Figure 9: KL divergence for beat types with start token 7729, demonstrating correct modeling of musical subdivisions.

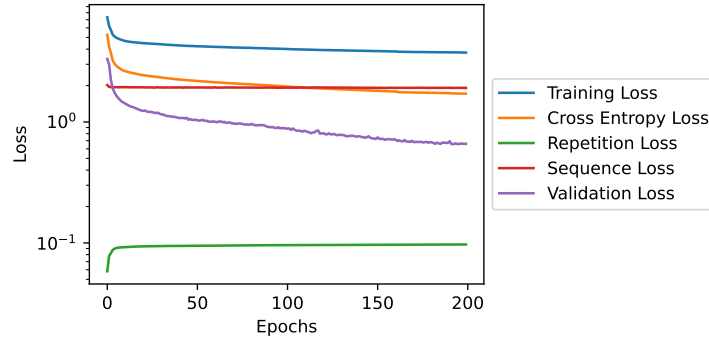


Figure 6: Loss plots over 200 epochs for a median of 3 independent training runs for hyper-parameter values listed in Table 4

3.2 KL DIVERGENCE EVALUATION

To quantitatively assess the model’s generative quality, the predicted token distributions were compared with the true distributions using KL divergence. Figures 7-9 visualize the results. Given that the model is autoregressive and expects an initial token, selecting the start token based on the distribution of starting notes of musical measures in the training data yields more consistent generation. The results were averaged over 10 generated phrases per prompt.

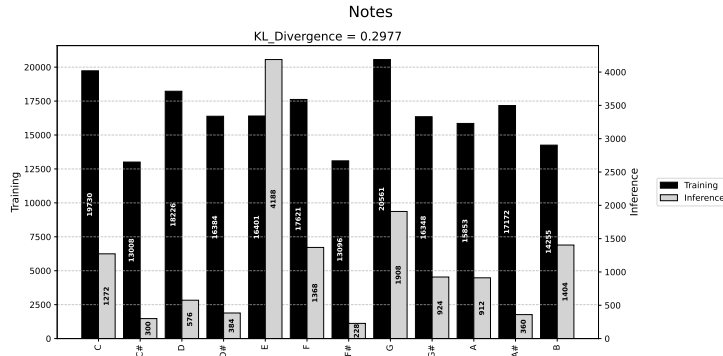


Figure 7: KL divergence between generated and true distributions of musical notes, indicating accurate note prediction by the model.

4 CONCLUSIONS

The proposed tokenisation scheme efficiently incorporates expressive elements without needing an explicit representation like chord vocabulary. The sequence-level attention mechanism captures phrase segmentation, enabling variable-length batching while preserving musical coherence. The model demonstrates strong performance in generating guitar riffs with expressive details, as evidenced by low KL divergence scores across pitch, beat, and beat type distributions. By incorporating a repetition penalty loss, the model naturally reduces redundancy without requiring explicit inference constraints. Future work could explore extending the approach to other instruments and incorporating additional musical attributes.



Figure 10: Five generated musical measures demonstrating the model’s ability to maintain musical coherence and expressive details at TEMPERATURE 0.7.

A TOKEN ID

Table 3: Token IDs for various accents.

Accent	Token ID
BAR	27049
BOS	27050
BARRE_NOTE	27051
BEND_NOTE_1	27052
BEND_NOTE_2	27053
BEND_NOTE_3	27054
BEND_NOTE_4	27055
BEND_NOTE_5	27056
BEND_NOTE_6	27057
BEND_NOTE_7	27058
TREM_BAR_1	27059
TREM_BAR_2	27060
TREM_BAR_3	27061
TREM_BAR_4	27062
TREM_BAR_5	27063
DEAD_NOTE	27064
SLIDE_NOTE_1	27065
SLIDE_NOTE_2	27066
SLIDE_NOTE_3	27068
SLIDE_NOTE_4	27069
SLIDE_NOTE_5	27070
SLIDE_NOTE_6	27071
HAMMER	27072
VIBRATO	27073
HARMONIC_1	27074
EOS	27075

B HYPER-PARAMETER TUNING

The following hyper-parameter values were used for training the model: The values were obtained

Table 4: Hyper-parameter values used for training.

Property	Value
FFN_HIDDEN	1024
MAX_SEQ_LENGTH	32
NUM_HEADS	4
DROP_PROB	0.3
NUM_LAYERS	2
D_MODEL	256
TEMPERATURE	1.0
OVERLAP	8
LAMBDA	1.5
ALPHA	0.1
DELTA	1.0

after performing a grid search for 10 epochs and selecting values which gave the lowest validation loss. The ranges for the grid search are given in the repository <https://github.com/DjgentleViBe/SCORE>.

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